

An open-source and experimentally guided CFD strategy for predicting air distribution in data centers with air-cooling

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ABSTRACT

Data centers are generally over cooled to ensure the trouble free running. In a data center with air-cooling, it is crucial to investigate the air distribution and temperature field to aid the design, which ensures the stable operation with less energy consumption. Computational fluid dynamics (CFD) is perfect for such a mission, but the cost is high and the user is offered with little customization as commercial software are mostly adopted. Therefore, this study presented a systematic investigation on making use of open-source software, including OpenFOAM and paraView to realize geometry preparation, mesh generation, experimentally guided numerical setup and solution, and results visualization. A JAVA program was developed to ensure the case preparation and simulation in just one command. Self-adapted momentum sources were developed to realize the desired flow rate through the servers. The strategy was validated and demonstrated by a pilot data center from Alibaba cloud, Alibaba group. The developed solver predicted the air temperature in both cold and hot isles of a data center with mean error of 0.7 K. This work initiated a starting point for achieving automated CFD simulation of data centers with open-source tools.

1. Introduction

A data center is a facility composed of networked computers, storage systems, and computing infrastructure for assembling, processing, storing, and disseminating large amounts of data. As our economy and society continue to shift towards digitalization, there is growing emphasize on cloud computing, data storage, and development of powerful servers [1]. Thus the energy consumption by data centers keeps increasing. The data centers in US consumed 1.8% of the total energy consumption in 2014 (estimated 70 billion kWh) [2]. Worldwide, data centers consumed 270 TWh in 2012 and the annual growth rate was 4.4% from 2007 to 2012 [3]. The resulted high economic and adverse environmental impact drives the crucial need for energy-efficient data centers [4].

A breakdown of energy consumption by a data center shows that the cooling system consumes 50%, servers 26%, power conversion 11%, and network 10% [5]. There are other studies show that the cooling system consumes 30%–45% of the total energy consumption in a data center [6,7]. To achieve an energy-efficient data center, on the one hand, chipset manufacturing companies strive to produce highly packed transistors to reduce the energy consumption by the servers.

On the other hand, deploying energy-efficient information technology (IT) equipment would reduce the capacity and energy consumption of the cooling infrastructure. However, the data centers are generally over cooled to ensure the trouble free running [7]. Therefore, the cooling system should be well designed and operated [8].

The cooling infrastructure generally consists of a refrigeration system and terminal cooling systems [9]. The terminal cooling systems could be air-cooling, liquid-cooling, and free-cooling. Air-cooling is most widely used technology in data centers due to its flexibility, simple maintenance, and acceptable operation cost [10–12]. An air-cooling system could be room level [13], row level [14], and rack level [15], depending on the location of the air handling units. The server racks divide the room into two spaces: cold aisle, the space on the front side of the servers and hot aisle, the space on the rear side of the servers. The major challenges in an air-cooling system is the cold air bypass, hot air recirculation, and negative pressure close the air handling unit. Such challenges would lead to hot spot and local high temperature. It is thus crucial to investigate the air distribution and temperature field in data centers to aid the design and operation of the air-cooling system, which ensures the stable operation of servers with less energy consumption.

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Nomenclature

α	porosity
β	coefficient of expansion
δ_{ij}	Kronecker delta
γ	relaxation factor
$\hat{n}$	1 if the air within a server flows in the x direction and otherwise -1
κ_{eff}	effective conductivity
ν_0	laminar kinematic viscosity
ν_t	turbulent kinematic viscosity
ν_{eff}	effective kinematic viscosity
ρ	density
ξ	mean relative error
A	cross section area of a server
c_p	specific heat capacity
d	length that air is accelerated in the x direction within a server
g_i	gravity component in the x_i direction
m	number of servers
n	level of grid refinement
P	power of a server
p	pressure
Pr_0	laminar Prandtl number
Pr_t	turbulent Prandtl number
Q	flow rate through a server
S_i	momentum source in the x_i direction
S_T	energy source
S_{U_i}	momentum source for a server in the x_i direction
sU_i	momentum source for a running server in the x_i direction
sU_x	momentum source for a running server in the x direction
T	temperature
T_{inlet}, T_{outlet}	inlet/outlet air temperature of a server
T_{ref}	reference temperature
U_i	mean velocity component in the x_i direction
$U_{x,cal}$	calculated velocity in the x direction within a server
$U_{x,tar}$	target velocity in the x direction within a server
V	volume of a server
x, y, z	coordinates
x_i	i th direction of coordinate, $i = 1, 2, 3$

The air distribution and temperature field in data centers could be obtained by solving the Navier–Stokes equations with computational fluid dynamics (CFD). With a validated turbulence model and accurate boundary conditions (BCs), CFD simulations could be reliable and informative. By reviewing the CFD studies in data centers with air-cooling during the last ten years in Table 1, commercial software such as ANSYS Fluent [61], STAR CCM+ [62], and 6SigmaRoom [30] are most popular in the research community. This might be due to the friendly user interface and comprehensive tutorials. However, the commercial software are highly packed and sometimes they are unable to cater the specific requirements in the simulation of data centers. For example, a user-defined function (UDF) is a function that one programs and can be dynamically loaded with the ANSYS Fluent to enhance the standard features of the code. But the implementation

Table 1

CFD studies in data centers with air-cooling.

Refs.	Year	Software	Type
[16]	2012	BlueTool based on OpenFOAM	publicly available
[17]	2012,	Thetis by I2M laboratory	N/A
[18–20]	2012	ANSYS Fluent	commercial
[21]	2013	Thetis by I2M laboratory	N/A
[22–26]	2013	ANSYS Fluent	commercial
[27]	2013	FloTHERM	commercial
[28]	2014	ANSYS Fluent	commercial
[29]	2014	BlueTool based on OpenFOAM	publicly available
[30,31]	2014	6SigmaRoom	commercial
[32]	2015	STAR CCM+	commercial
[33,34]	2015	6SigmaRoom	commercial
[35]	2015	FloTHERM	commercial
[36]	2015	FloVENT	commercial
[37–39]	2016	ANSYS Fluent	commercial
[40]	2017	ANSYS Fluent	commercial
[41,42]	2017	6SigmaRoom	commercial
[43]	2018	STAR CCM+	commercial
[44,45]	2018	ANSYS Fluent	commercial
[46,47]	2019	6SigmaRoom	commercial
[48]	2019	ANSYS Fluent	commercial
[49,50]	2019	Research code in C++	N/A
[51]	2020	6SigmaRoom & ANSYS Fluent	commercial
[52]	2020	Research code in C++	N/A
[53–55]	2020	ANSYS Fluent	commercial
[56]	2021	FloTHERM	commercial
[57]	2021	Autodesk CFD	commercial
[58]	2022	OpenFOAM	open-source
[59]	2022	ANSYS Fluent	commercial
[60]	2022	STAR CCM+	commercial

of a new algorithm or the solution of a new continuity function is not feasible. Besides, the use of commercial software would increase the operation cost. In contrast, open-source software generally has no good tutorial and nice user graphic interface, but gives the user full freedom in implementing one's own methodologies and they are free. There are few studies on open-source software and public platforms. BlueTool is an open platform framework and publicly available through the Web site: <http://impact.asu.edu/BlueTool/> [16]. This platform was developed based on the open-source CFD software OpenFOAM [63]. But this web site is not accessible and there is no related publications since 2014. There was one recent study that investigated OpenFOAM for the efficiency assessment of data centers [58]. Their study showed the feasibility of OpenFOAM in modeling the IT equipment. The momentum of servers was modeled by assigning a constant velocity to the servers' domain in each iteration, which is not feasible for a large data center with numbers of servers running independently. Besides, a complete and usable open-source solution asks more, including pre-processing, post-processing, automation, and the capability of using the on-site monitoring data or being “experimentally guided”.

Therefore, this study presented a systematic investigation on making use of open-source softwares to realize geometry preparation, mesh generation, experimentally guided numerical setup and solution, and results visualization. Self-adapted momentum sources were developed to realize the desired flow rate through the servers. Please note that such a source is quite hard to implement in commercial codes based on the authors' experience. The strategy was validated and demonstrated by a pilot data center from Alibaba cloud, Alibaba group. This work initiated a starting point for achieving automated CFD simulation of data centers with open-source tools.

2. Methodology

2.1. Numerical model

The turbulent flow in a data center could be regarded as incompressible with the Boussinesq approximation [64] to include the buoyancy

Table 2
Server information and state.

RackID	A02	A02	A03	...
UPos	4	9	18	...
UNum	1	2	1	...
Model	XX1	XX2	XX3	...
Time	2022/8/9 1:45	2022/8/9 1:45	2022/8/9 1:00	...
T_{inlet} (°C)	24	23	23	...
T_{outlet} (°C)	34		35	...
Power (W)	304	159	299	...
Fan1 (rpm)	10950	4128	11880	...
Fan2 (rpm)	9600	4320	9600	...
Fan3 (rpm)	11100	4128		...
Fan4 (rpm)	9450	4320		...

effect. Then the governing equations are:

$$\frac{\partial U_i}{\partial x_i} = 0 \quad (1)$$

$$U_j \frac{\partial U_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_i} \left\{ \nu_{eff} \left[\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) - \frac{2}{3} \left(\frac{\partial U_k}{\partial x_k} \right) \delta_{ij} \right] \right\} + S_i \quad (2)$$

$$U_j \frac{\partial T}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\kappa_{eff} \frac{\partial T}{\partial x_j} \right) + S_T \quad (3)$$

where U_i is the mean velocity component (m/s) in the x_i direction, ρ the air density (kg/m³), p the air pressure (Pa), $\nu_{eff} = \nu_0$ (laminar air kinematic viscosity) + ν_t (turbulent air kinematic viscosity) the effective kinematic viscosity (m²/s), δ_{ij} the Kronecker delta, T the air temperature (K), and $\kappa_{eff} = \nu_0 / \text{Pr}_0 + \nu_t / \text{Pr}_t$ the effective conductivity (m²/s). Pr_0 and Pr_t are laminar and turbulent Prandtl numbers, respectively. The turbulent air kinematic viscosity is calculated with the re-normalization group (RNG) $k - \epsilon$ model [65] that has good performance in built environment applications [66].

The solution of the Navier–Stokes equations adopted Semi-Implicit Method for Pressure Linked Equations (SIMPLE) algorithm [67] to couple the pressure and velocity. The scalar equations for parameters k , ϵ , and T were solved iteratively with an implicit scheme. The convection terms are discretized with a bounded second-order upwind scheme.

2.1.1. Momentum source

The source term in Eq. (2) is constructed by the Boussinesq approximation for buoyancy force and an explicit momentum source S_U for IT equipment:

$$S_i = g_i [1 - \beta(T - T_{ref})] + S_{Ui} \quad (4)$$

where g_i the gravity component in the x_i direction (m²/s), β the coefficient of expansion (K⁻¹), T_{ref} the reference temperature (K), and S_{Ui} the momentum source for the servers that is constructed by:

$$S_{Ui} = \begin{cases} sU_i & \text{if server is on} \\ \alpha U_i & \text{if server is off} \end{cases} \quad (5)$$

If a server is on, a momentum source sU_i is added. If a server is off, the server domain is assigned an individual porosity α that is modeled by Darcy's law [68]. This study made use of on-site monitored data in Table 2 if a server is on, then the simulation was experimentally guided. Table 2 gives the server information and monitored temperatures, power, and fan speed. The RackID (index of a rack) and UPos (index of a server unit in the rack) determined the location of the server and UNum determines the height of a server unit. Then the exact location and dimensions of this server was obtained.

The momentum source sU_i was sU_x in x direction and zero in other directions in this study. The source worked as an acceleration, and its strength was initialized by treating the air accelerated from zero to a target velocity $U_{x,tar}$ in the x direction:

$$sU_x = \frac{U_{x,tar}^2}{2d} \frac{U_{x,tar}}{|U_{x,tar}|} \quad (6)$$

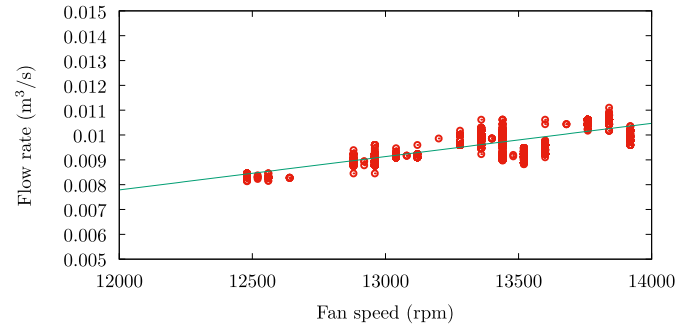


Fig. 1. Flow rate (m³/s) vs. fan speed (rpm).

where $U_{x,tar}$ was the target velocity, d the length that air was accelerated in the x direction within a server, and $|U_{x,tar}|$ the norm of the velocity.

For the servers whose inlet/outlet air temperature were collected in Table 2, the target velocity could be calculated by:

$$U_{x,tar} = \hat{n} \frac{P}{\rho c_p A (T_{outlet} - T_{inlet})} \quad (7)$$

where P was the power of the server (W), c_p the air specific heat capacity (J/(kgK)), A the cross section area of the server (m²), T_{outlet} the outlet air temperature of the server, and T_{inlet} the inlet air temperature. $\hat{n}$ defines the direction of the velocity, which was 1 if the air within a server flows in the x direction and otherwise -1 . However, the temperature data collected in the server room was incomplete. Therefore, this investigation made use of the relationship between the flow rate and fan speed. By using the existing data, the relationship could be fitted into a line by $Q = a \sum rpm_i + b$, where Q was the flow rate (m³/s). Fig. 1 gives an example. Then the target velocity can be calculated by:

$$U_{x,tar} = \hat{n} \frac{Q}{A} \quad (8)$$

This initial momentum source was obtained by assuming the air velocity at the server entrance being zero, which was not true with the existence of air conditioners. Therefore, the momentum needed to be updated to realize the target velocity. With a calculated air velocity $U_{x,cal}$ in the x direction within a server, the dynamic momentum source was updated by:

$$sU_x^{new} = sU_x^{old} + \gamma \left(\frac{\frac{U_{x,tar}}{|U_{x,tar}|} U_{x,tar}^2 - \frac{U_{x,cal}}{|U_{x,cal}|} U_{x,cal}^2}{2d} \right) \quad (9)$$

where γ was a relaxation factor that used to stabilize the computation.

2.1.2. Heat source

The source strength (K/s) S_T in Eq. (3) was assigned by:

$$S_T = \frac{P}{\rho c_p V} \quad (10)$$

where P was the power of the server (W) that could be obtained from Table 2, c_p the air specific heat capacity (J/kgK), and V the volume of a server (m³).

2.2. Automated simulation

A JAVA program named Automatic Initialization of Data center simulation (AID) was developed to generate OpenFOAM case files. It can automatically read geometric structures from STL files and arrange them for mesh generation. Source terms and BCs can also be generated from the server information and state in Table 2. The numerical setups used those in Section 2.1 as default. It will further generate an script with a Python file to automatically create the mesh, run the simulation,

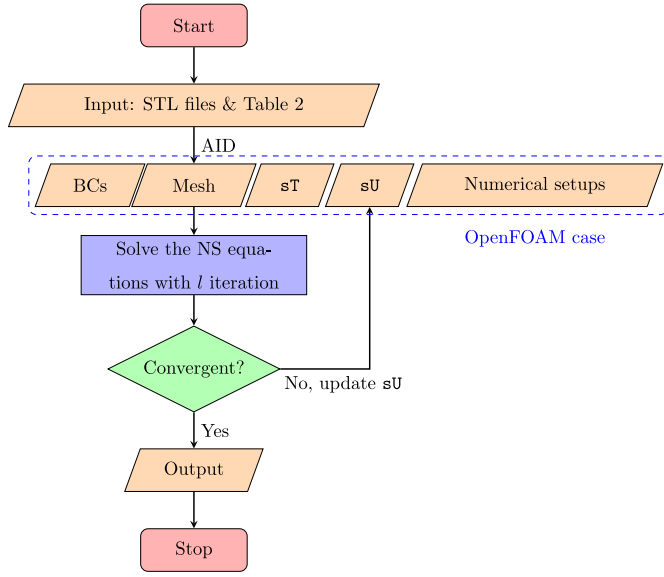


Fig. 2. Solution flow chart.

update sU, and collect data in one command. The AID allows a more accessible and user-friendly CFD tool for the numerical simulation of data centers.

Fig. 2 shows the solution flow chart that starts by creating the OpenFOAM case files with STL files for geometry and Table 2 for the servers as inputs. In the case files, mesh was generated and source terms, BCs, and numerical setups were all set. This study ran the steady-state simulation with l iteration before checking convergence. The convergence criterion were twofold. First, the residuals for energy less than 10^{-6} and those for other variables less than 10^{-3} . Second, the target air velocity at the servers $U_{x,tar}$ was reached by checking:

$$\sum_{m=1}^M \frac{|U_{x,tar} - U_{x,cal}|}{|U_{x,tar}|} / m \leq \xi \quad (11)$$

where m was the number of servers and $\xi = 10\%$ in this study.

2.2.1. Geometry preparation and mesh generation

Alibaba cloud uses building information modeling (BIM) files with Industry Foundation Classes (IFC) extension. IFC is a popular format used in building and construction industry that provides interoperability between different software applications. Therefore, many software supports the import and export of IFC format files, whether they are open source or closed source. Further, the structure of a IFC file not only facilitates users to understand the hierarchy but also is convenient for computer to identify and process data.

In this study, the BIM software Autodesk Revit [69] exports the geometry of a data center to standard triangle language (STL) files. A STL file stores information about the triangular facets that tile the object surface with either ASCII encoding or binary encoding. The information of each triangle includes the coordinates of the vertices and the normal unit vector to the triangle. Fig. 3 shows such a model of a data center with 170 racks and each rack has 44 units of servers. The STL models may need simplifications if there are too many details. Firstly, the structures such as ducts that has little influence on the accuracy of simulations are removed. Then the structures such as the gaps between a pillar and a rack that requires extremely fine mesh but again has little impact on the simulations are modified. Finally, if two objects have shared surfaces, the surface should not be duplicated. This study adopted an open-source software Blender 3.2 [70] to simplify the geometry. Blender can directly edit points, edges, and faces of the triangular faces, so it is feasible to simplify the model.

Based on simplified STL files, OpenFOAM is able to generate a structured hexahedral mesh by the utility `snappyHexMesh`. `snappyHexMesh` starts with `surfaceFeatureExtract` and `blockMesh`. `surfaceFeatureExtract` with default parameters is utilized to extract features from the model files. Then a background mesh that completely wraps the model is created via `blockMesh`. The aspect ratio of all grids are suggested to be 1:1:1. The grid size should ensure at least one intersection of a cell edge with the STL surfaces. The background grid size was set to be 0.8 m in this study after grid independence test. Based on the extracted features and background mesh, the user needs to tell where to refine the mesh and the corresponding refinement level. Besides the boundaries, one can create refinement boxes for specific spaces, e.g. refinement boxes for the racks were created in this investigation. Then `castellatedMeshControls` defines refinement levels for all the boundaries and refinement boxes. n levels refinement will reduce the grid size to $(1/2)^n$ of the background grid size. The refinement boxes are given a refinement level 4, the inlet and outlet level 3, and other geometries level 2. The default `snapControls` ensure that the mesh snaps to the model surfaces. One can use `addLayersControls` to grow grid layers on specified boundaries. Since all the important boundaries have been well refined in this study, there is no need to use `addLayersControls`.

2.2.2. Source terms for a server

The momentum sources S_{U_i} for active servers were defined by creating a `volVectorField` sU within the server domain. The domain for a momentum source was defined by `box (x1 y1 z1) (x2 y2 z2)` from the server information in Table 2. Then $d = x2 - x1$ is the length of the source in the x direction. sU is initialized with `setFieldsDict` to assign the source strength in the domains and zero in other spaces. An example is:

```

regions
(
    boxToCell
    {
        box (x1 y1 z1) (x2 y2 z2);
        fieldValues (volVectorFieldValue sU (sU_x 0 0));
    }
)
  
```

For an inactive server, there was no heat generation, but there was airflow through it. Therefore, the corresponding domain should be modeled as porous zone. A porosity field was defined with `volScalarField` alpha for α in Eq. (5). Since the inactive servers in this study were blocked completely, alpha was initialized with `setFieldsDict` to assign the porosity 1 000 for the inactive servers and zero for else spaces. An example for the definition is:

```

regions
(
    boxToCell
    {
        box (x1 y1 z1) (x2 y2 z2);
        fieldValues (volScalarFieldValue alpha 1000);
    }
)
  
```

To setup the heat dissipated by the servers, this study created a `volScalarField` sT that explicitly represents the heat sources S_T . sT was initialized with `setFieldsDict` to assign the source strength in the servers' domain and zero in other spaces. An example for the definition is:

```

regions
(
    boxToCell
  
```

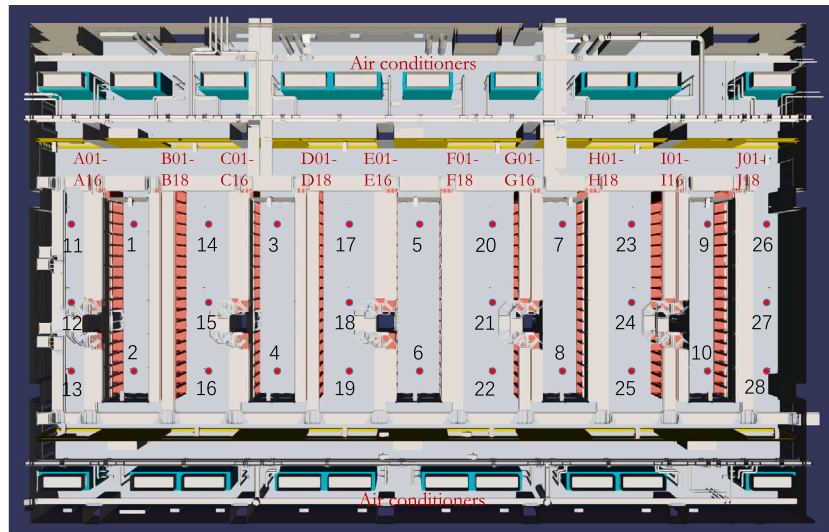


Fig. 3. Geometry of a data center from BIM, the red dots represent temperature sensors.

```
{
  box (x1 y1 z1) (x2 y2 z2);
  fieldValue (volScalarFieldValue sT  $\frac{P}{\rho c_p V}$ );
}
```

The domain of a heat source, e.g. the server, was again defined by box (x1 y1 z1) (x2 y2 z2) from the server information in Table 2. Then V the volume of the heat source (m^3) was defined by (x1-x2) (y1-y2) (z1-z2).

2.3. Results visualization

Visualization was done through an open-source software paraview [71]. It can render geometry in a variety of ways, such as surface and surface with mesh. The result fields can be plot as contour maps or vector maps. Most importantly, it offered a wide variety of data analysis methods, such as slicing, vector plots, flow charts, or plotting simulation results for probe locations.

2.4. Data acquisition

To validate the numerical simulation, this study collected data from the data center shown in Fig. 3. Since the data center was in operation, the working state was changing and further the thermal state. Fig. 4 shows the monitored total power in 1440 min (24 h). The sampling interval was 15 min. Since it was necessary to use the steady-state data for validation, this investigation used the data between 180 and 195 min. The power data for individual servers were also checked and the time selected was that most of the servers had little power fluctuations.

Besides the data showed in Table 2, the data center also measured the air temperature at the hot aisles and cold aisles as marked by red dots in Fig. 3. The measurements used E+E Elektronik EE210 humidity and temperature sensor with an error of ± 0.2 K to collect temperature data every 2.5 min. The CFD simulation was validated by comparing the predicted air temperature with the measured ones.

3. Results

3.1. Geometry and mesh

Fig. 3 shows the geometry of a data center exported from BIM. There were lots of geometric details such as ducts, pipes, and gaps, etc. that

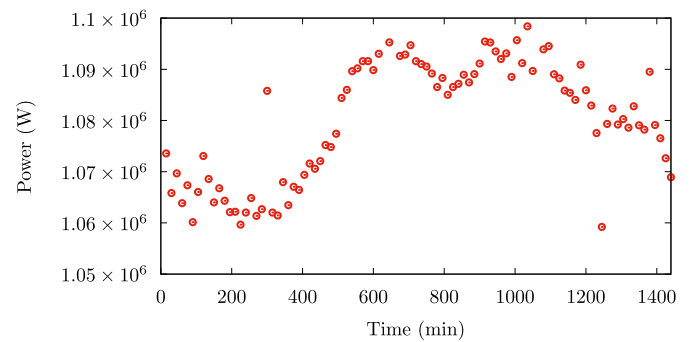


Fig. 4. Monitored total server power.

requires fine mesh but had little impact on the accuracy of predictions in the target region. Fig. 5 shows the simplified geometry that would greatly reduce the number of grid in mesh generation. The two rows of air conditioners were located on the two sides of the room. There were ten columns of servers, from A to J. In one column, there were either 16 or 18 racks. The spaces between A and B, C and D, E and F, G and H, and I and J were the hot aisles and the rests were cold aisles. The servers and the air conditioners were separated by internal walls with openings that help lead the supply air to cold aisles. The air was exhausted from the top of the hot aisles.

Based on the simplified geometry, this study conducted grid independence test. The results led to a mesh with 3 200 000 cells. Fig. 6(a) shows a horizontal cross section. The mesh at the servers, air conditioners and close-to-wall region are refined. Fig. 6(b) further shows the grid distribution for racks A11-A16, the refinement level 4 was applied to the region inside the racks, so that each server could be accurately modeled.

3.2. CFD solutions

This study ran the simulation with $l = 500$ iteration, $\gamma = 1.0$, and $\xi = 10\%$ for convergence. It means that for every 500 iteration, the program updated the momentum source for each server based on Eq. (9). This was a trade-off between having sufficient iteration before updating the momentum sources and having more updates in limited total iteration to obtain fast convergence. Fig. 7 shows the variation of momentum source sU_x and calculated x velocity component $U_{x,cal}$ with respect to iteration for server C07-9 as an example. The dash line represents the

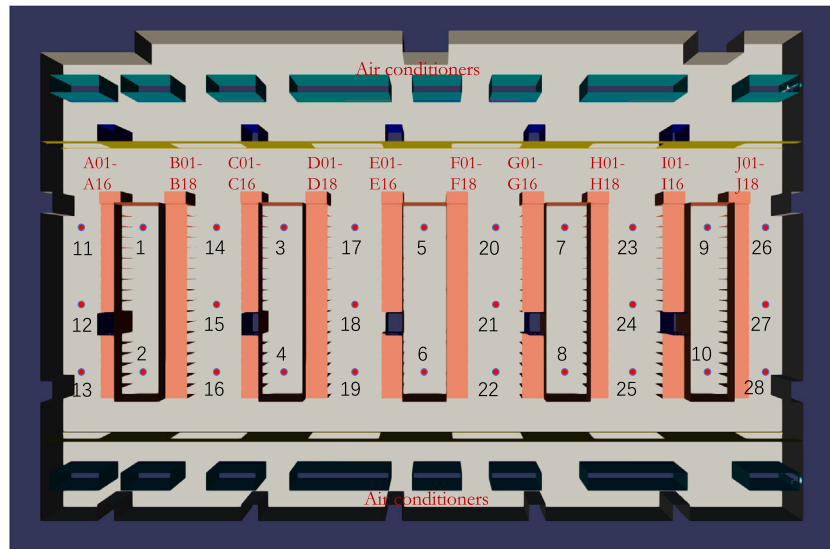


Fig. 5. Simplified geometry of a data center, the red dots represent temperature sensors located 2.2 m above the floor. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

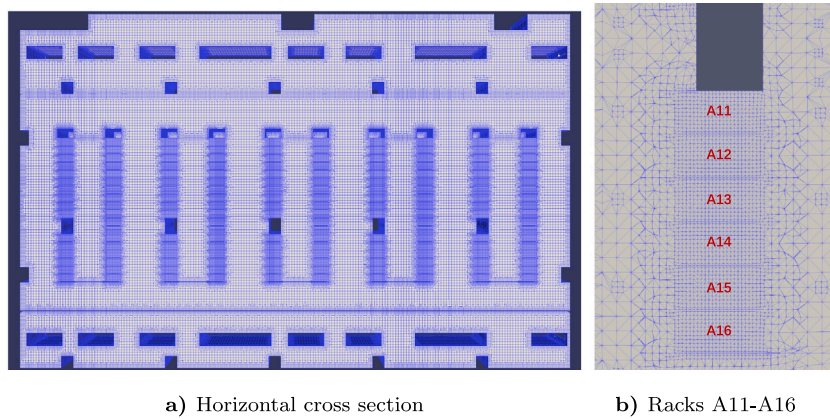


Fig. 6. Mesh for the data center.

target air velocity in the x direction $U_{x, tar}$. The convergence could be reached after updating the momentum sources for 70 times or 35 000 iteration. In the end of the computation, the sU_x and $U_{x, cal}$ showed minor variations that implied convergence.

Fig. 8 shows the flow field at $z = 0.5$ m, the plane at which intersects both the servers and the air supply grille of the air conditioners. Please note that the internal wall between the servers and air conditioners helped guide the air to the cold aisles. With the momentum sources, there was almost no reversed flow at the servers. This agreed with the actual condition and our expectation. The predicted temperature field at $z = 0.5$ m in Fig. 9 further confirmed that warm air in the hot aisles can hardly enter the cold aisles. There was unusual high temperature at servers in racks H06, H07, and H08. These servers were off and there was no air movement. The high temperature was due to the heat transfer from those servers above and below.

For validation, this study compared the predicted air temperature with the monitored ones in Fig. 10. Those locations refer to the numbered points in Fig. 3 and they were 2.2 m above the floor. The experimental data were averaged values from the 15-min measurements. The mean temperature error was 0.7 K that implied acceptable prediction accuracy. The well temperature agreement at the cold aisles further confirmed that the CFD simulation had little reversed flow. This was because the reversed flow would significantly increase the air temperature at the cold aisles. The greatest temperature difference

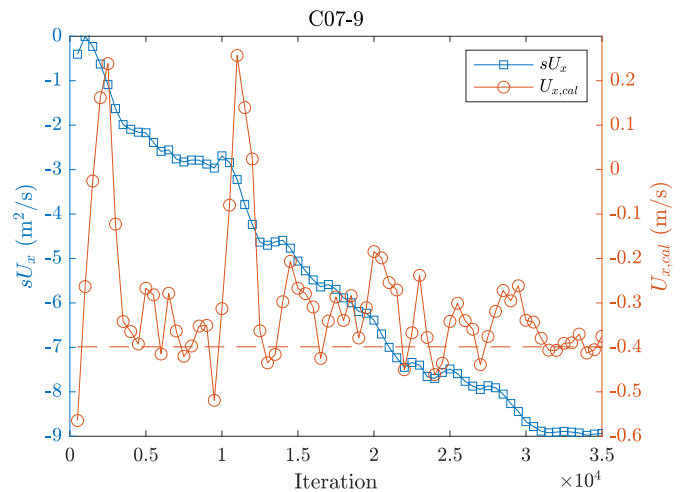


Fig. 7. Momentum source and air velocity vs. iteration for server C07-9.



Fig. 8. Predicted flow field at $z = 1.5$ m.

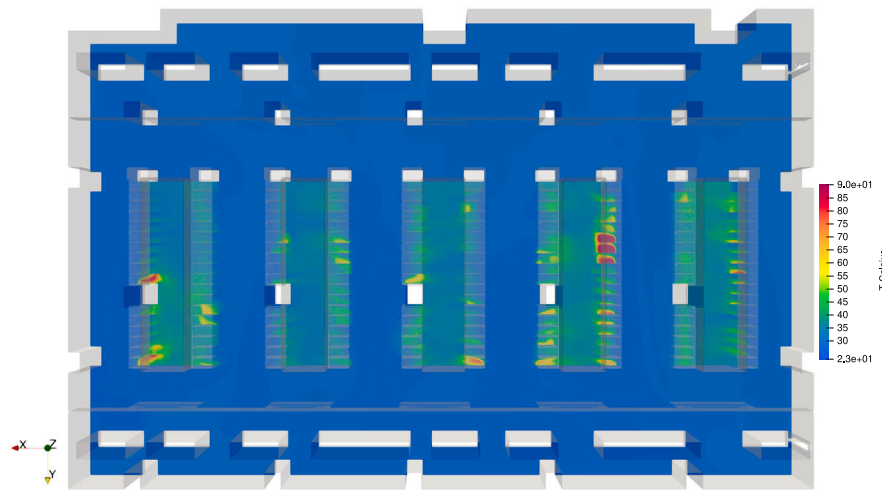


Fig. 9. Predicted temperature field at $z = 1.5$ m.

was 2.9 K that was at position 9. A possible reason was the inaccurate input of the servers' power or flow rate. Considering the complexity and capacity of the data center, the accuracy of the developed solver was acceptable.

4. Discussions

The automation of the developed tools were not perfect since the preparation of the geometry files still needs manual operations. The reason was that the exported STL files from BIM had too many details to be directly used in CFD simulations. To make up, one way is to take the CFD needs into consideration and develop more modularized BIM models. Another way could be development of a tool that takes the major geometric information such as dimensions and coordinates as input and outputs the STL files.

There are mainly three methods to simulate the momentum of a server. The first one is fixed velocity, which assigns a fixed velocity in the server in each iteration. This method works if you only need to fix the velocity in limited positions. In our case, there were thousands of servers. This study tried to fix the velocity at the servers but it could not lead to a converged solution. The second method is to make use of the fan curves to give a pressure increase at the servers. The accuracy of this method is dependent on the fan curves. There are many fans in a server, simulating each fan is too computational demanding. The

third method is to add momentum sources. If the pressure increase within a server could be determined, then the momentum source is fixed. The pressure increase of a single fan in a server could be obtained by using the rpm and fan curve. However, a server may have numbers of fan running in parallel or in serial. With the block of the server's electronic components, the combined effect on pressure increase is unknown. It is still a challenge to determine the pressure increase of a server. This study tried this method and there were significant reversed flow through some servers. Because the input data only determined the flow rate at each server, the proposed self-adapted momentum source ensured the desired airflow rate at the servers without contradicting the governing equations.

Please note that commercial software can also change the source terms like Eqs. (4) and (5) if the terms were constant. In this study, the momentum source sU_i in Eq. (5) is self-adapted to realize the desired airflow rate at the server by using Eq. (9). This adaption requires data conversion and processing between the CFD program and the Python code, which is quite difficult in commercial software. Besides, the default source definition in OpenFOAM or other commercial software requires the definition of source zones first and then the value. Such an approach has good generalizability, but the simulation would be greatly slowed down if there are thousands of sources, such as in a data center. This study thus created scalar fields for source terms so that they were directly embedded in the momentum and energy equations. Such an

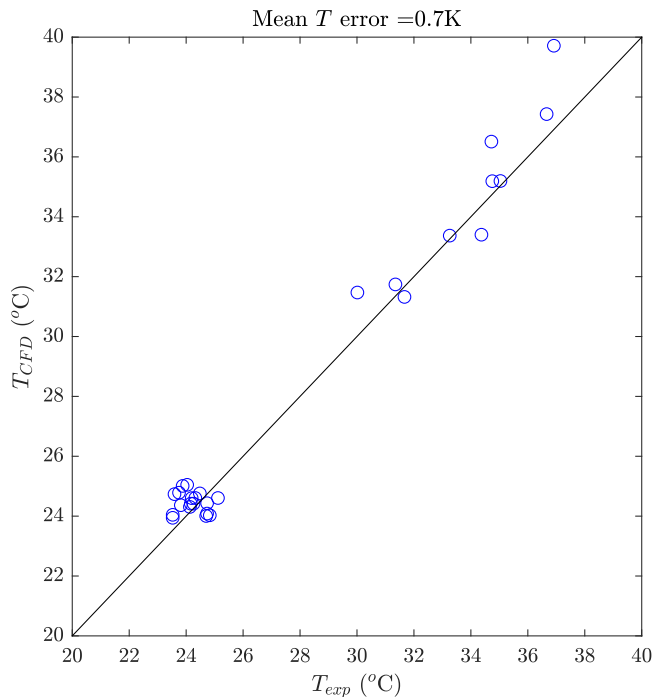


Fig. 10. Comparison on the predicted and measured air temperatures.

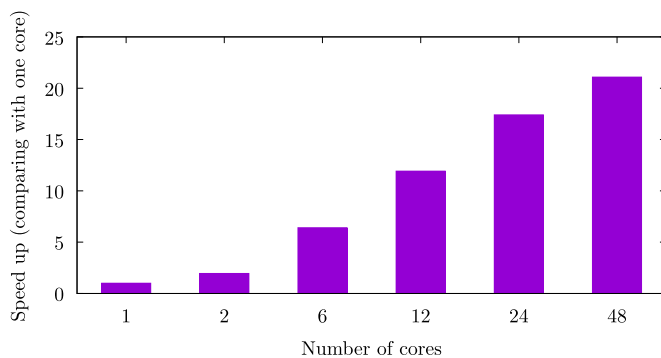


Fig. 11. Parallel speed up by comparing with one core performance.

approach did not require extra computation time for dealing with the source terms.

Another great advantage of this open source CFD program is the free parallel license. This study tested the parallel speed up in Fig. 11. The computation used a server with two sockets and each socket had an Intel Xeon Platinum 8269CY CPU with 26 cores running at 2.5 GHz. With 48 cores running in parallel, the speed up comparing with one core could be more than 20. The running time of the proposed method is dependent on the number of times in updating the momentum sources. Once the momentum sources are updated, the simulation requires hundreds or thousands of iteration for a converged solution.

The proposed strategy at current stage could be utilized for designing the cooling system of data centers. For control purpose, the real-time monitoring of servers' status could be used for predicting the velocity and temperature distribution, but not in real-time. The real-time predictions may possibly be realized by using machine learning or reduced-order model such as proper orthogonal decomposition analysis.

5. Conclusions

This study presented a systematic study on making use of open-source software to realize pre-process, numerical simulation, and post-process for CFD simulations of the air distribution and temperature fields in data centers. Due to the limited information on the operation status of the servers, self-adapted momentum sources were developed to realize the desired flow rate through the servers. With the input of geometries in STL format and server's operation status in tables, this study developed a JAVA program named AID that prepared the case files and run the simulations in one command. The strategy was validated and demonstrated by a pilot data center from Alibaba cloud, Alibaba group. The developed solver predicted the airflow with no evident reversed flow in the servers and air temperature with mean error of 0.7 K. This work proves the possibility of developing automated and open-source CFD tool for the numerical simulation of data centers.

CRedit authorship contribution statement

Wei Liu: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Conceptualization. **Song Lian:** Writing – original draft, Visualization, Software, Methodology. **Xin Fang:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Zhenyu Shang:** Writing – original draft, Conceptualization. **Hao Wu:** Visualization, Software. **Hao Zhu:** Writing – review & editing, Supervision, Funding acquisition. **Simon Hu:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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